9.3.1 Probability Mass Function.

If, X is a discrete random variable taking values $x_1, x_2 \ldots x_n$ with respective probabilities $p(x_1), p(x_2) \ldots p(x_n)$ such that

(i) $p(x_i) \geq 0$, $\forall$ i (non- negative) and (ii) $\sum_{i=1}^{n} p(x_i) = 1$ then $p(x)$ is known as the probability mass function (p.m.f) of the discrete random variable X.

The pair $\{x_i, p(x_i); i = 1, 2, 3, \ldots\}$ is known as prabability distribution of X.

Example 9.1

A coin is tossed two times. If X is the number of heads, find the probability mass function of X.

Solution:

Since the coin is tossed two times, the sample space is S={HH, HT, TH, TT}

If X denotes the numbers of heads, the possible values of X are 0,1,2 with the following

$P(X = 0) = P$ (getting no head) $= \frac{1}{4}$

$P(X = 1) = P$ (getting one head) $= \frac{2}{4} = \frac{1}{2}$

$P(X = 2) = P$ (getting two heads) $= \frac{1}{4}$

NOTE

Probabilities are non negative and the total is 1.

The probability distribution of X is

X	0	1	2
$p(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Example 9.2

In example 9.3 the probability mass function of X is given in the following table

X	2	3	4	5	6	7	8	9	10	11	12
$P(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The above table may be called as the probability distribution function of X.

9.3.2 Probability Density Function

When the random variable is continuous in the co-domain it is spread over it. A function $f(x)$ is defined on real line and satisfying the following conditions :